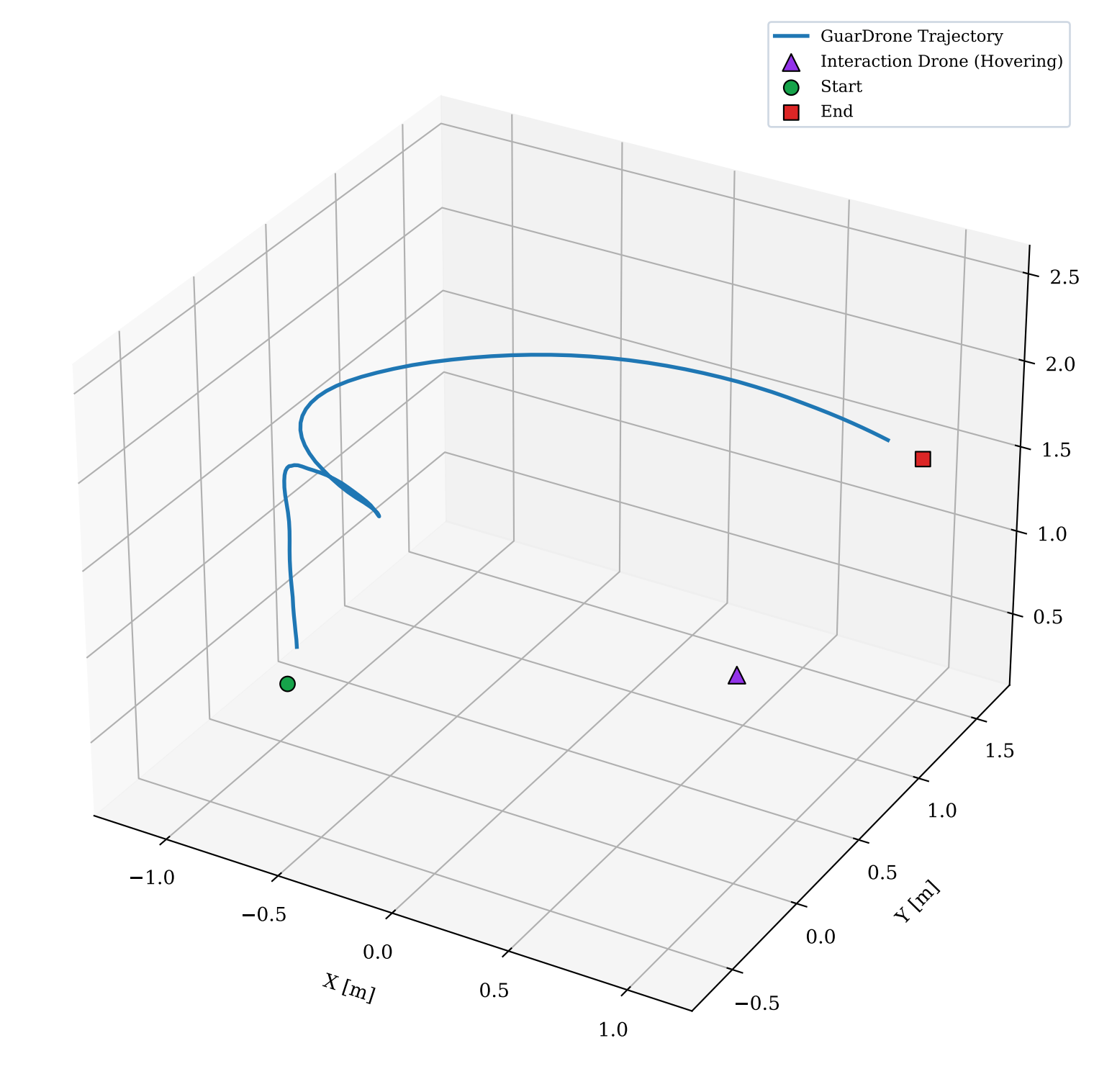
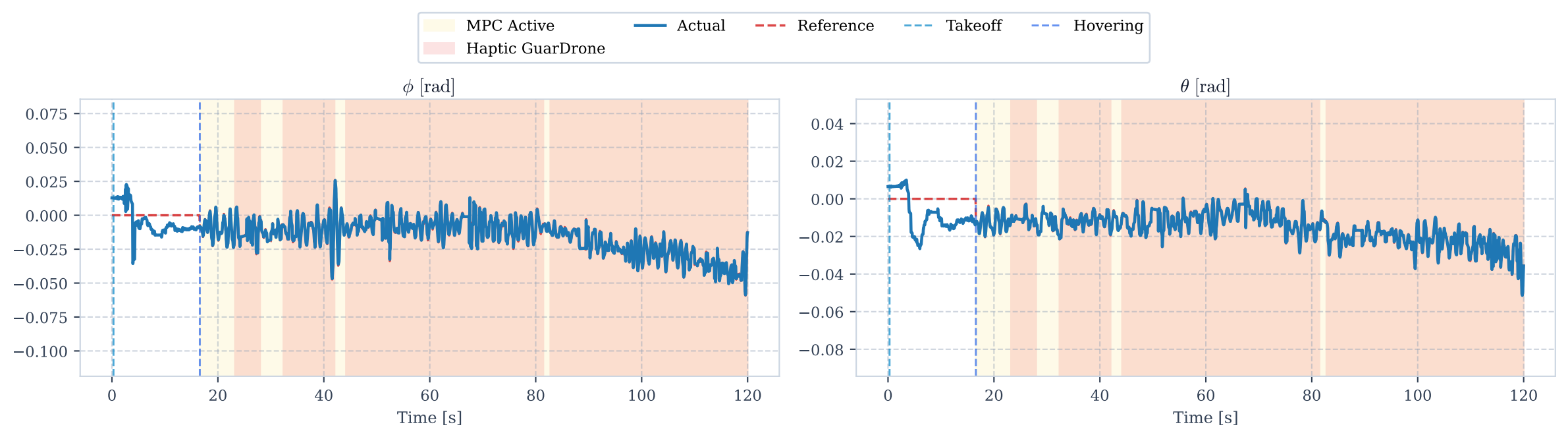
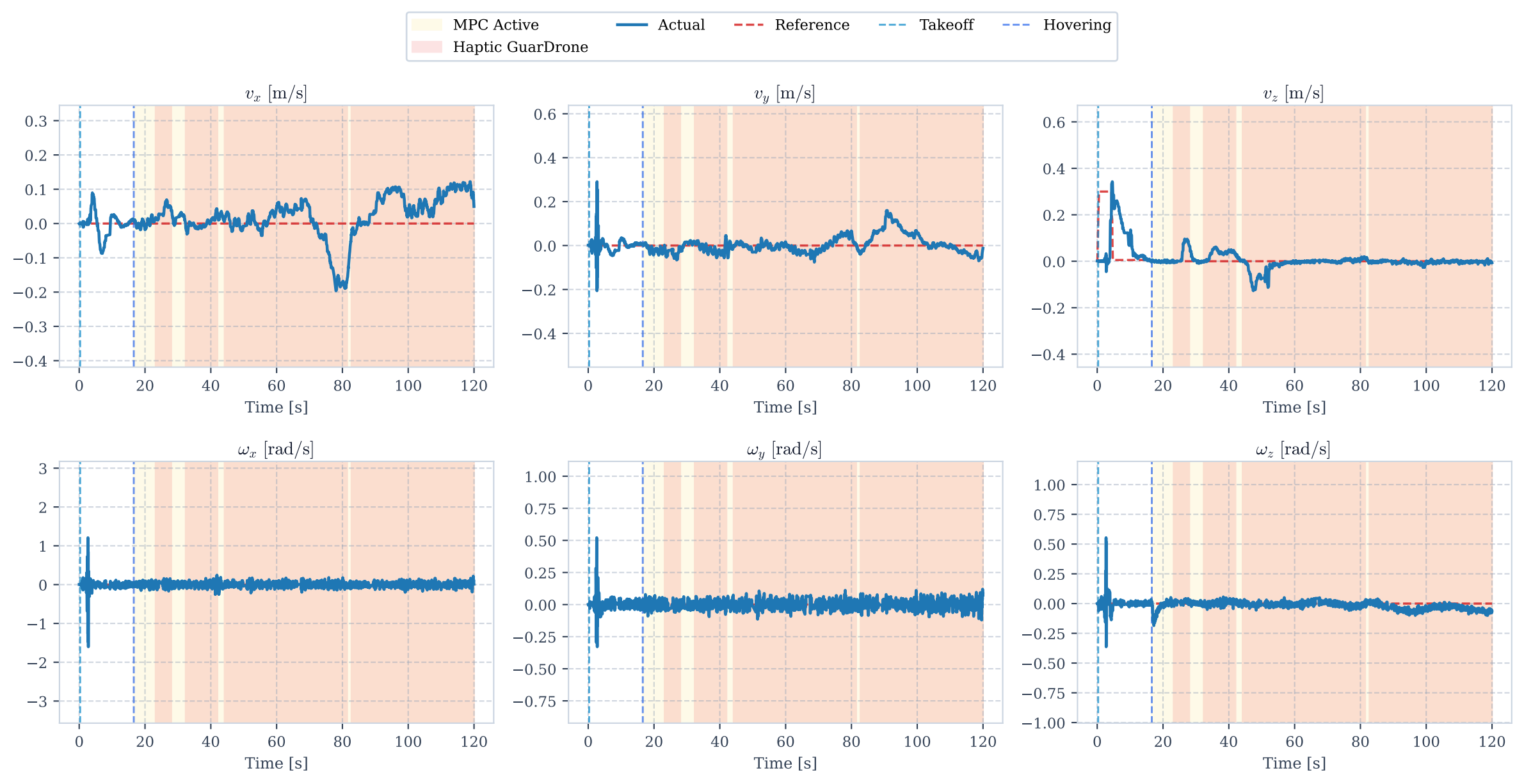
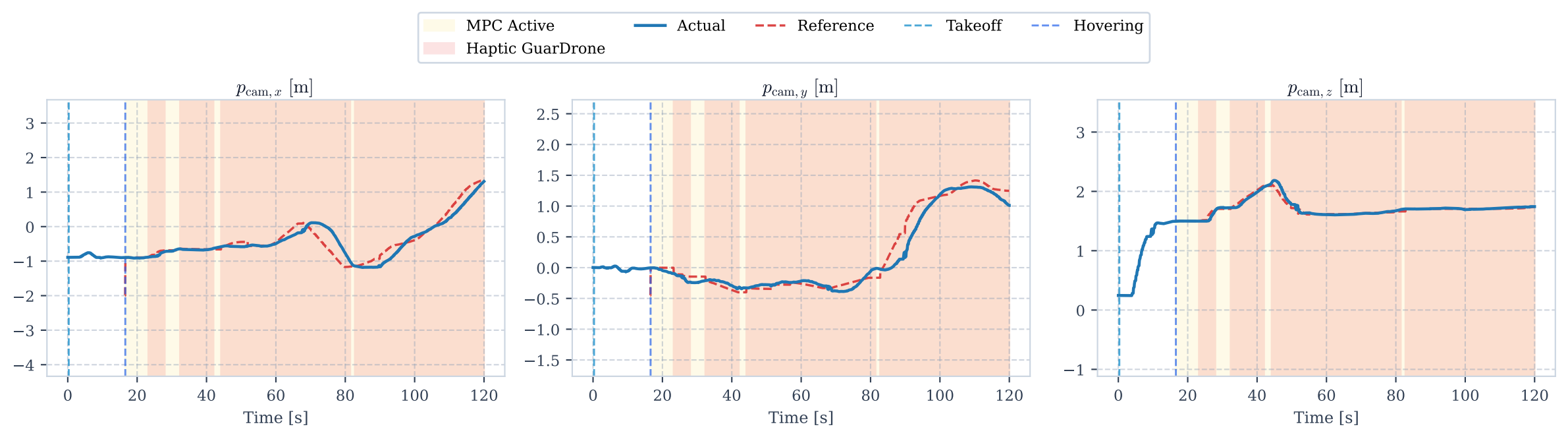
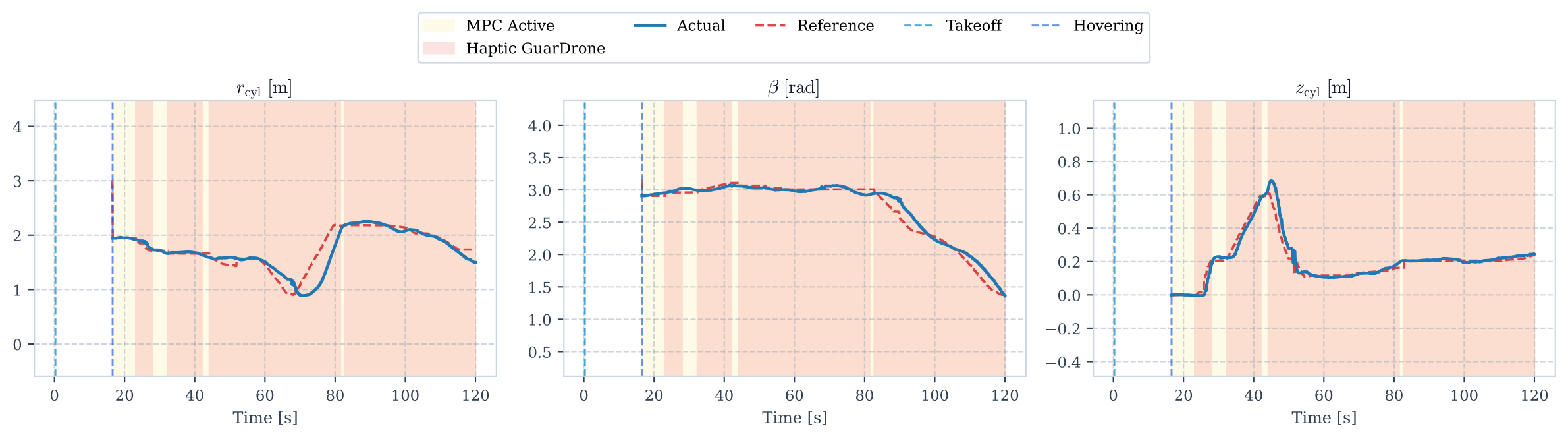
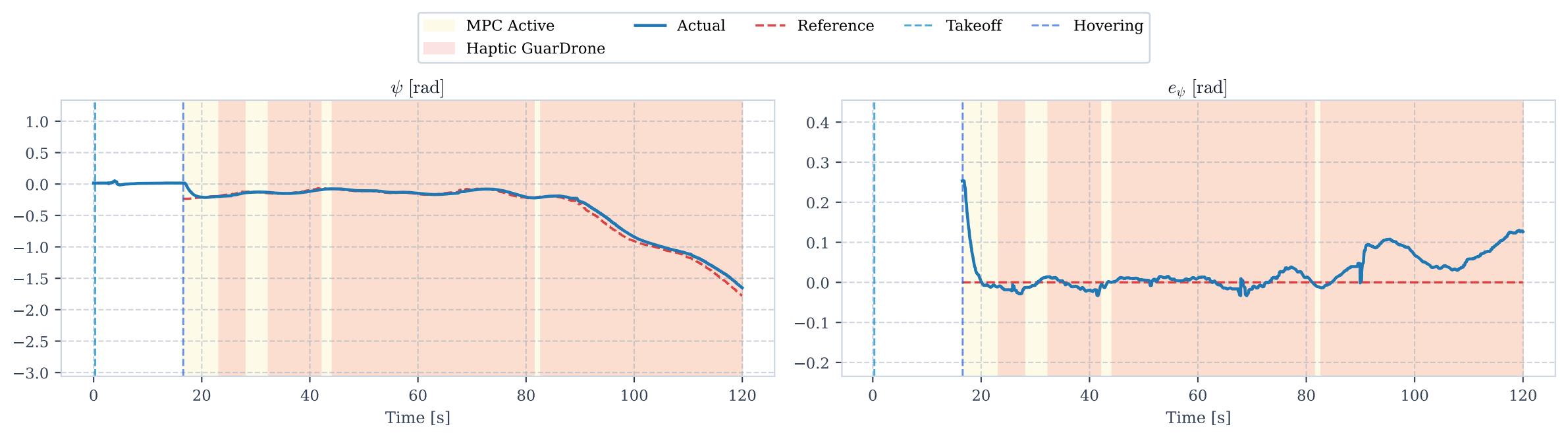

GuarDrone Trajectory
Interaction Drone (Hovering)
Start
End
X [m]
Y [m]
−1.0
−0.5
0.0
0.5
1.0
−0.5
0.0
0.5
1.0
1.5
0.5
1.0
1.5
2.0
2.5

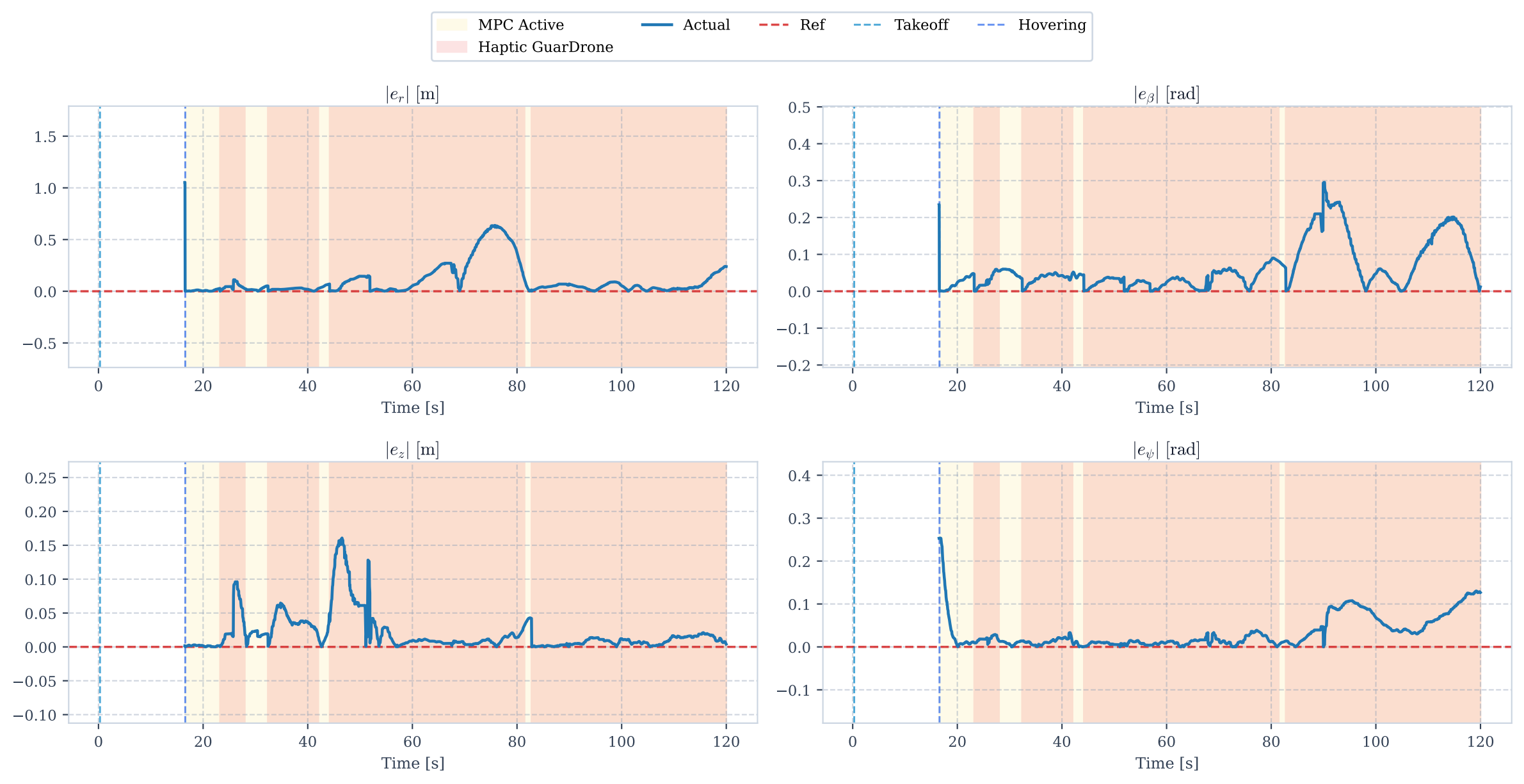

MPC Active
Haptic GuarDrone
Actual
Reference
Takeoff
Hovering
ϕ [rad]
θ [rad]
Time [s]
Time [s]

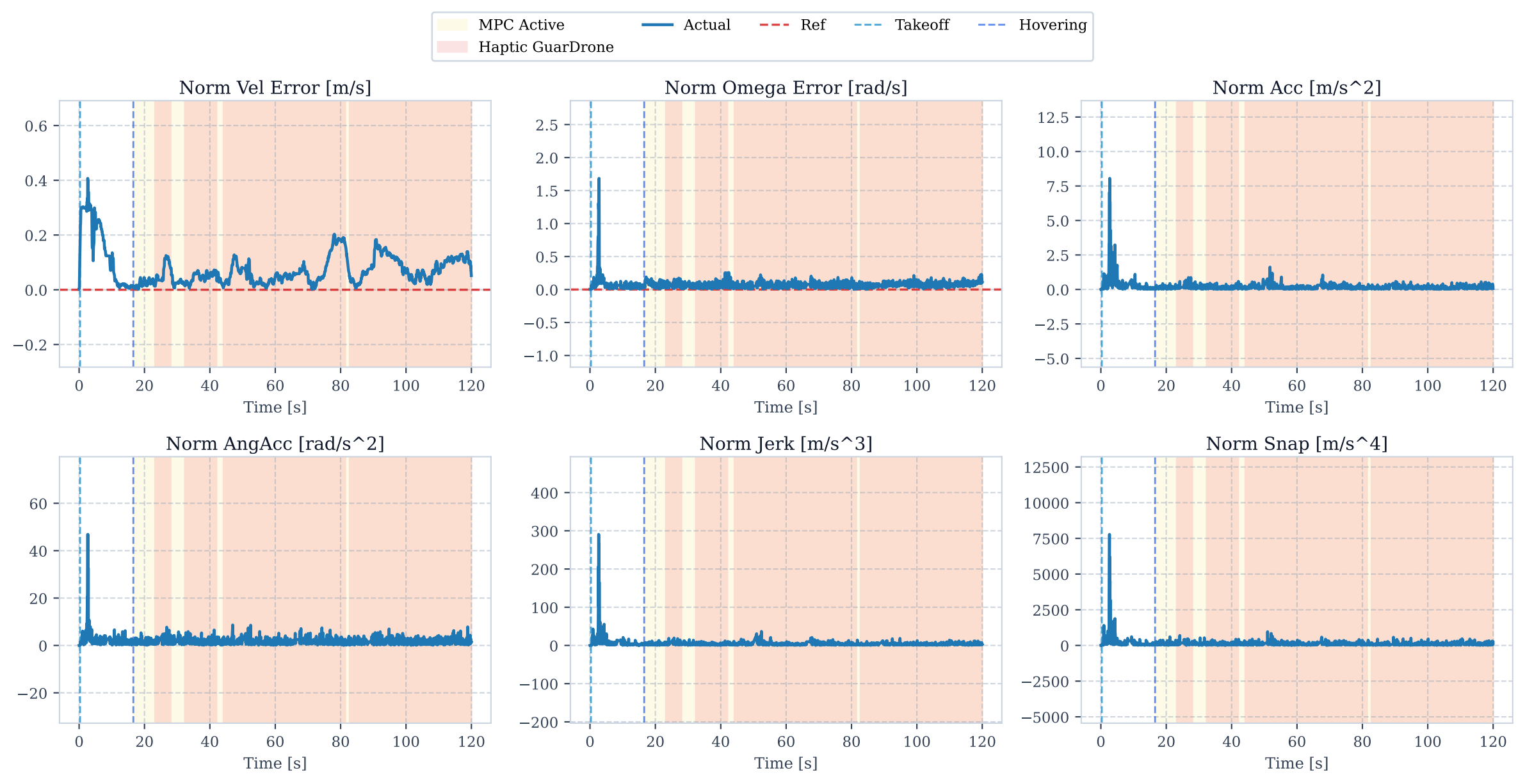

MPC Active
Haptic GuarDrone
Actual
Reference
Takeoff
Hovering
v_x [m/s]
v_y [m/s]
v_z [m/s]
ω_x [rad/s]
ω_y [rad/s]
ω_z [rad/s]
Time [s]

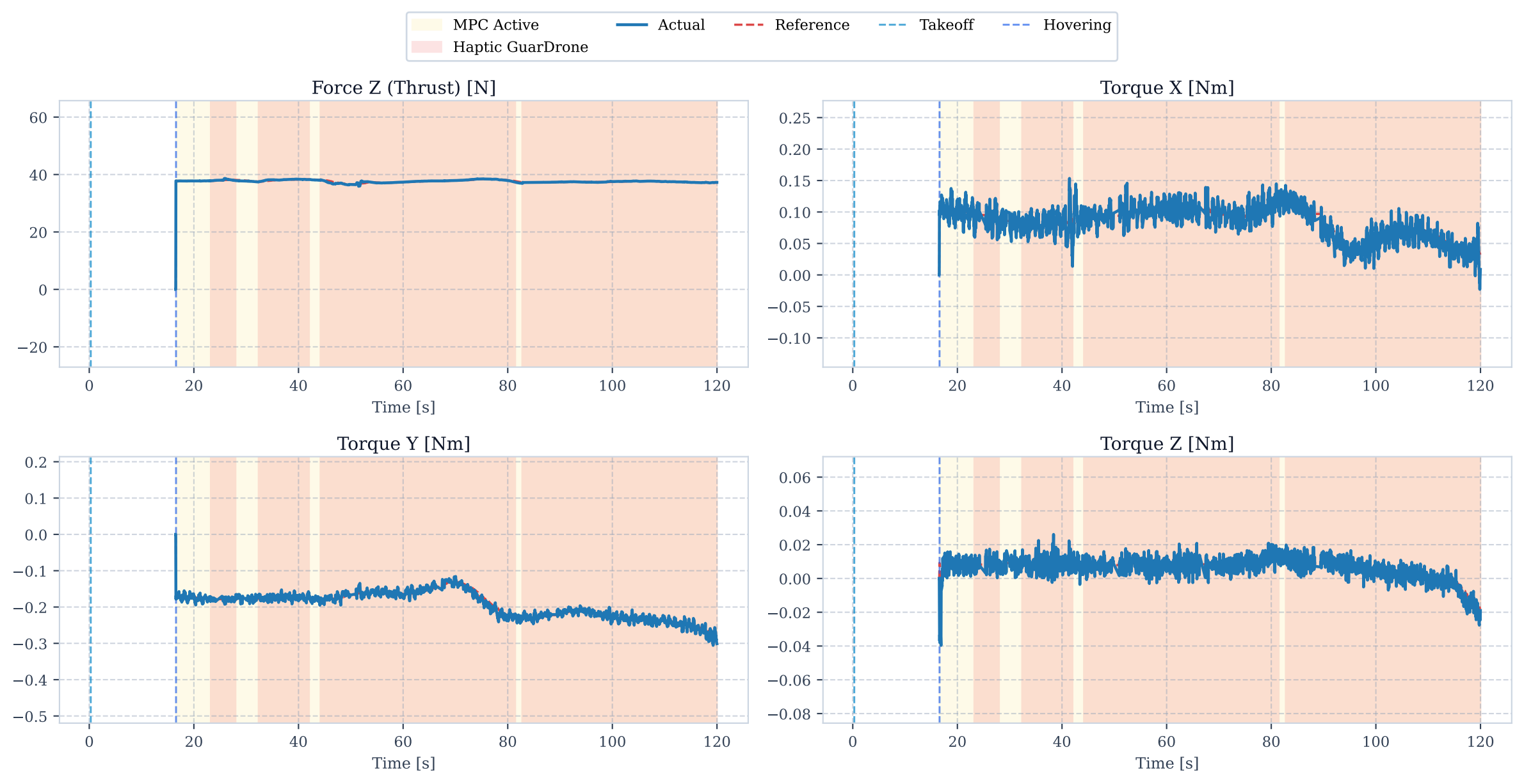

MPC Active
Haptic GuarDrone
Actual
Reference
Takeoff
Hovering
$p_{\text{cam},x}$ [m]
$p_{\text{cam},y}$ [m]
$p_{\text{cam},z}$ [m]
Time [s]
Time [s]
Time [s]

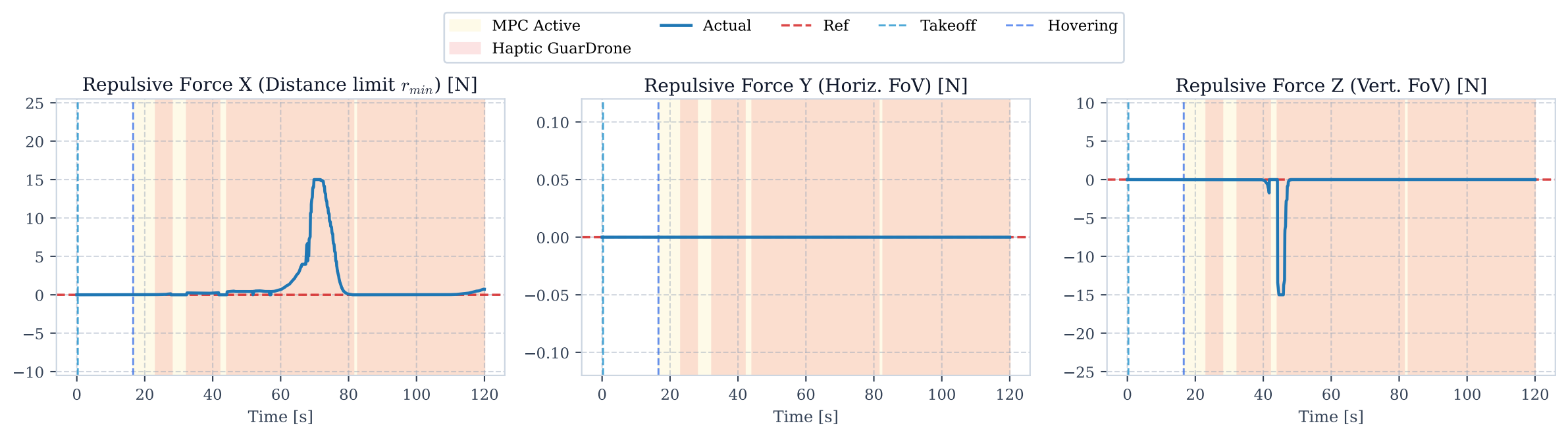

MPC Active
Haptic GuarDrone
Actual
Reference
Takeoff
Hovering
r_{cyl} [m]
β [rad]
z_{cyl} [m]
Time [s]
Time [s]
Time [s]

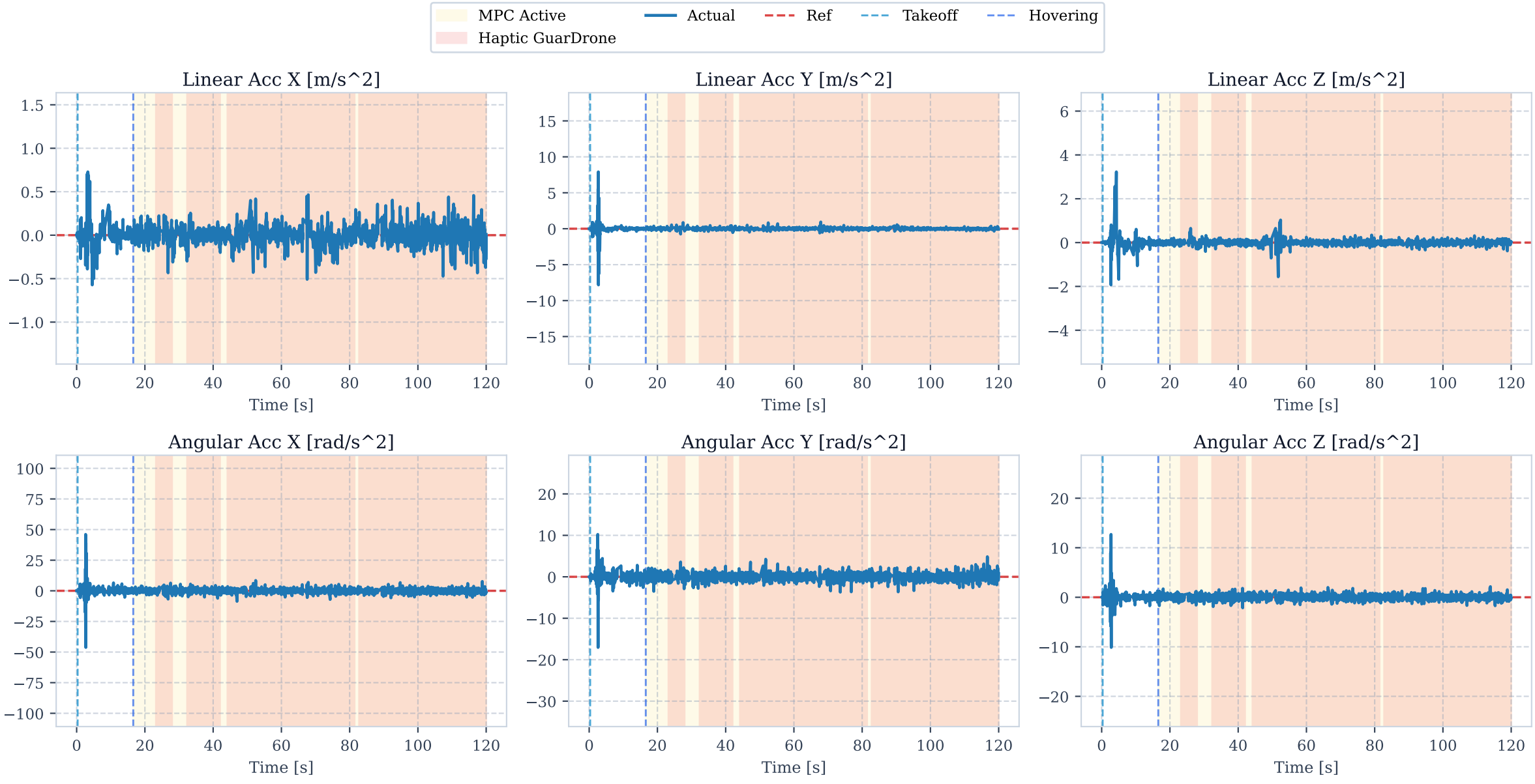

MPC Active
Haptic GuarDrone
Actual
Reference
Takeoff
Hovering
ψ [rad]
e_ψ [rad]
Time [s]
Time [s]

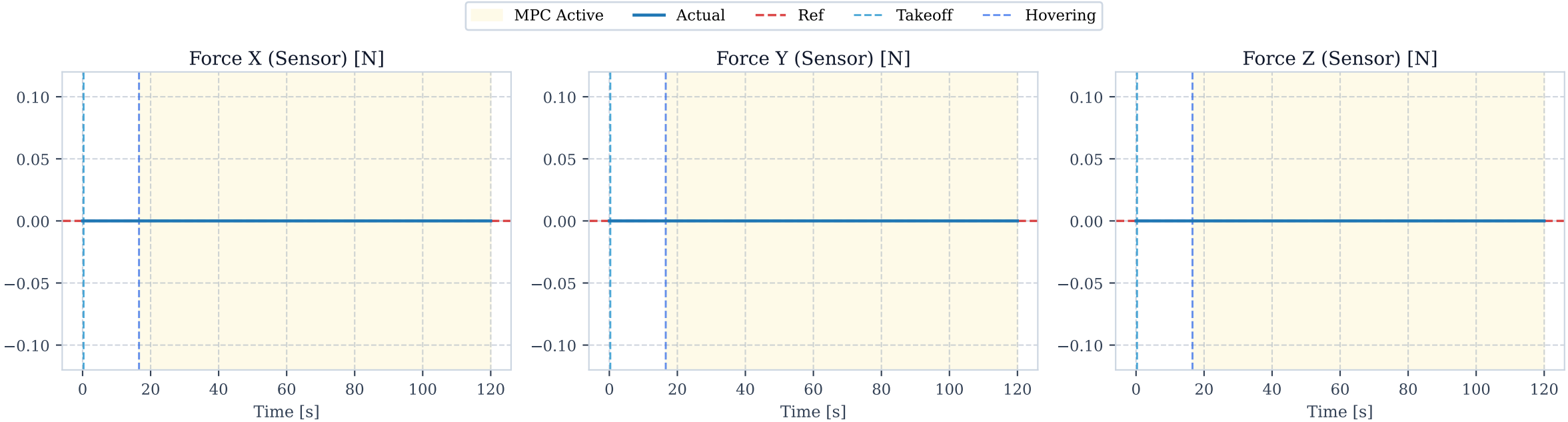

MPC Active
Haptic GuarDrone
Actual
Ref
Takeoff
Hovering
$|e_r|$ [m]
$|e_\beta|$ [rad]
$|e_z|$ [m]
$|e_\psi|$ [rad]
Time [s]
Time [s]
Time [s]
Time [s]

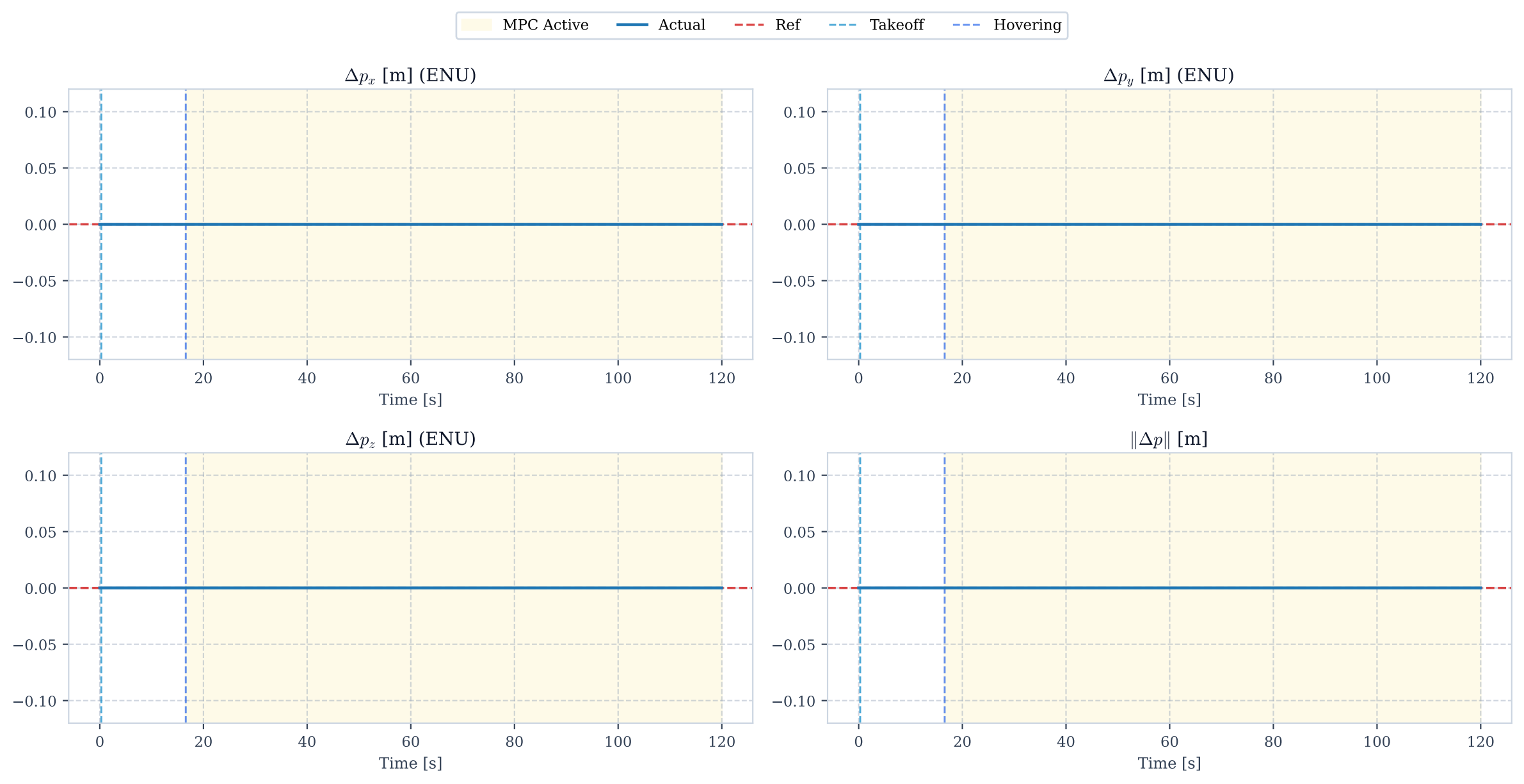

MPC Active
Haptic GuarDrone
Actual
Ref
Takeoff
Hovering
Norm Vel Error [m/s]
Norm Omega Error [rad/s]
Norm Acc [m/s^2]
Norm AngAcc [rad/s^2]
Norm Jerk [m/s^3]
Norm Snap [m/s^4]
Time [s]

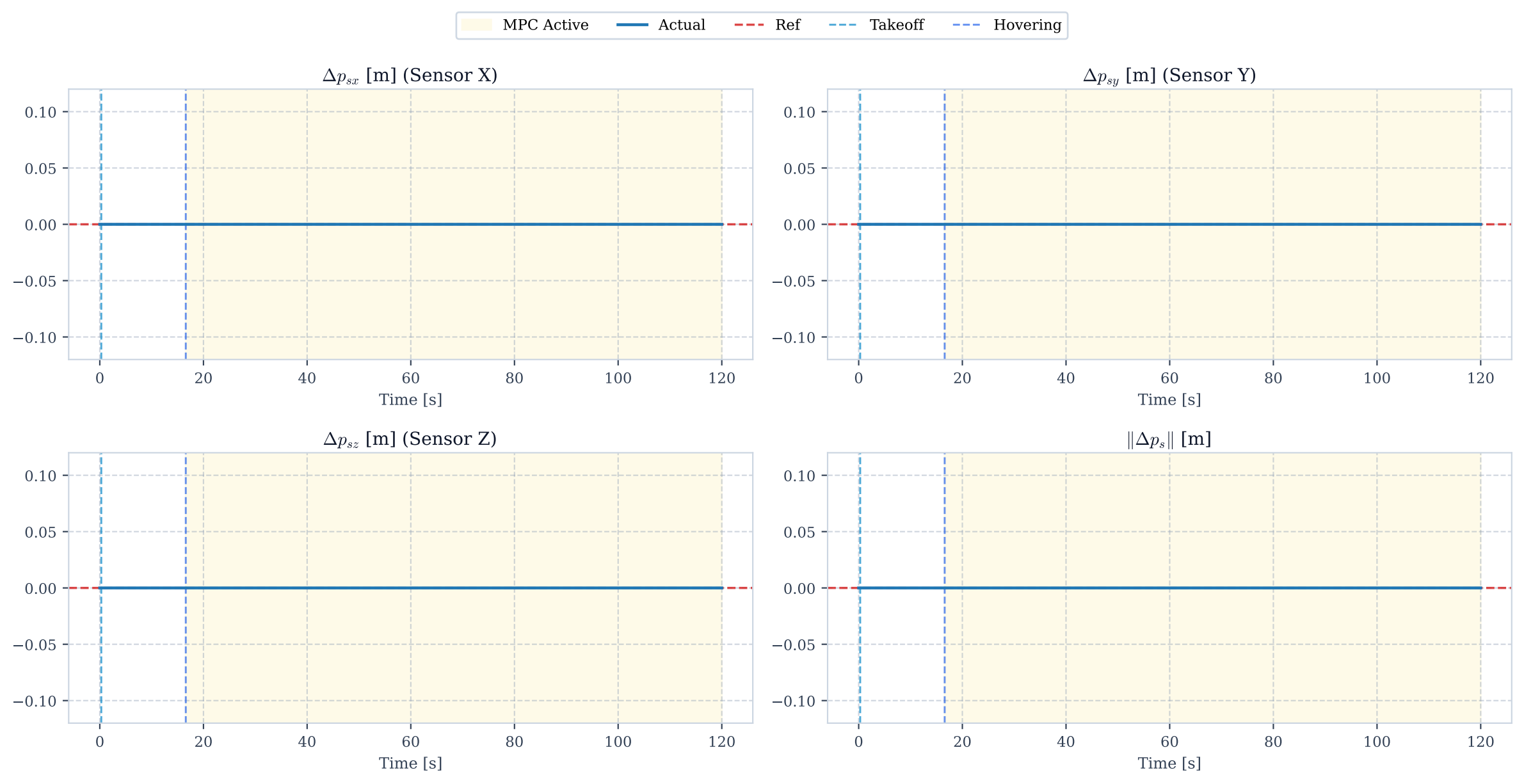

MPC Active
Haptic GuarDrone
Actual
Reference
Takeoff
Hovering
Force Z (Thrust) [N]
Torque X [Nm]
Torque Y [Nm]
Torque Z [Nm]
Time [s]
Time [s]
Time [s]
Time [s]

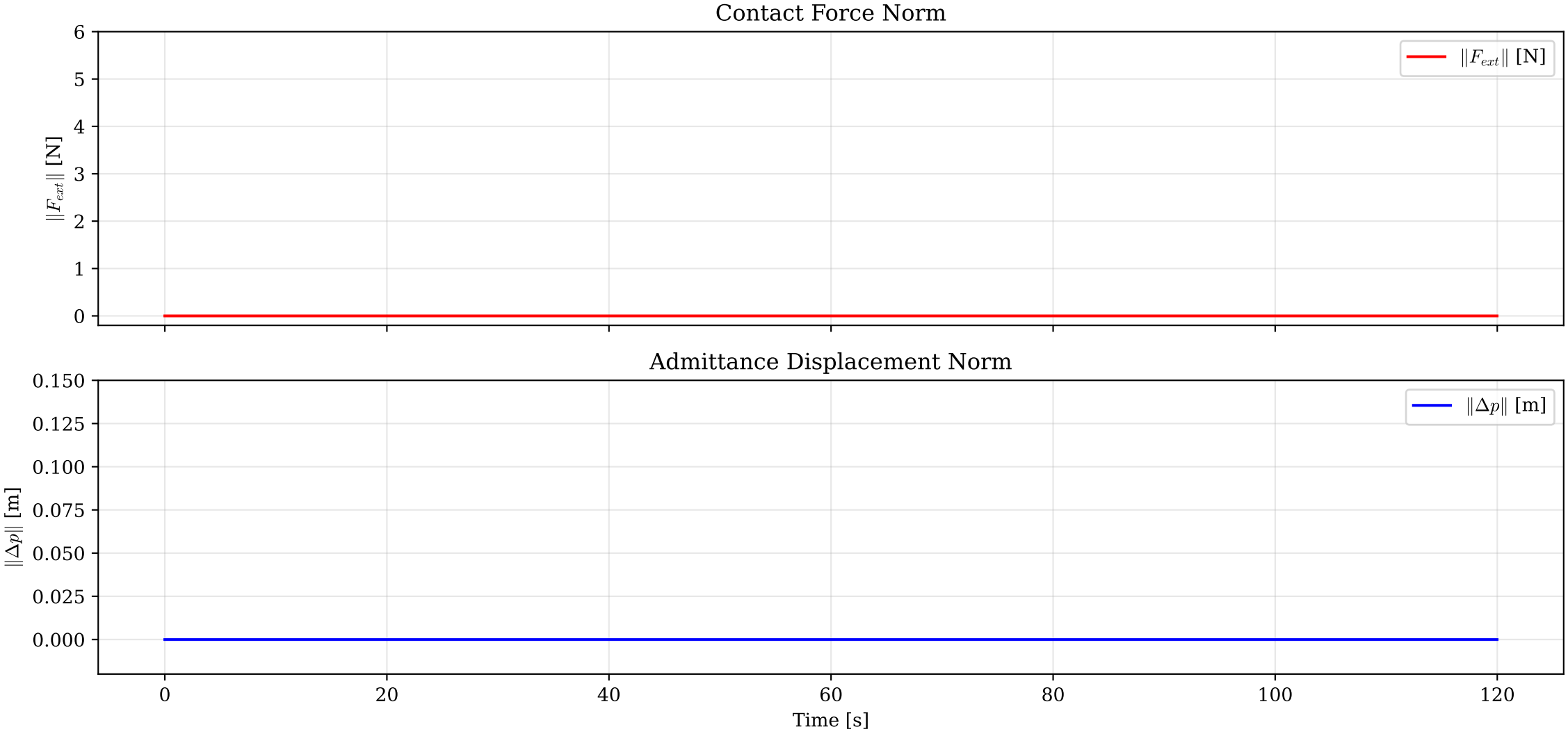

MPC Active
Haptic GuarDrone
Actual
Ref
Takeoff
Hovering
Repulsive Force X (Distance limit r_{min}) [N]
Repulsive Force Y (Horiz. FoV) [N]
Repulsive Force Z (Vert. FoV) [N]
Time [s]
Time [s]
Time [s]

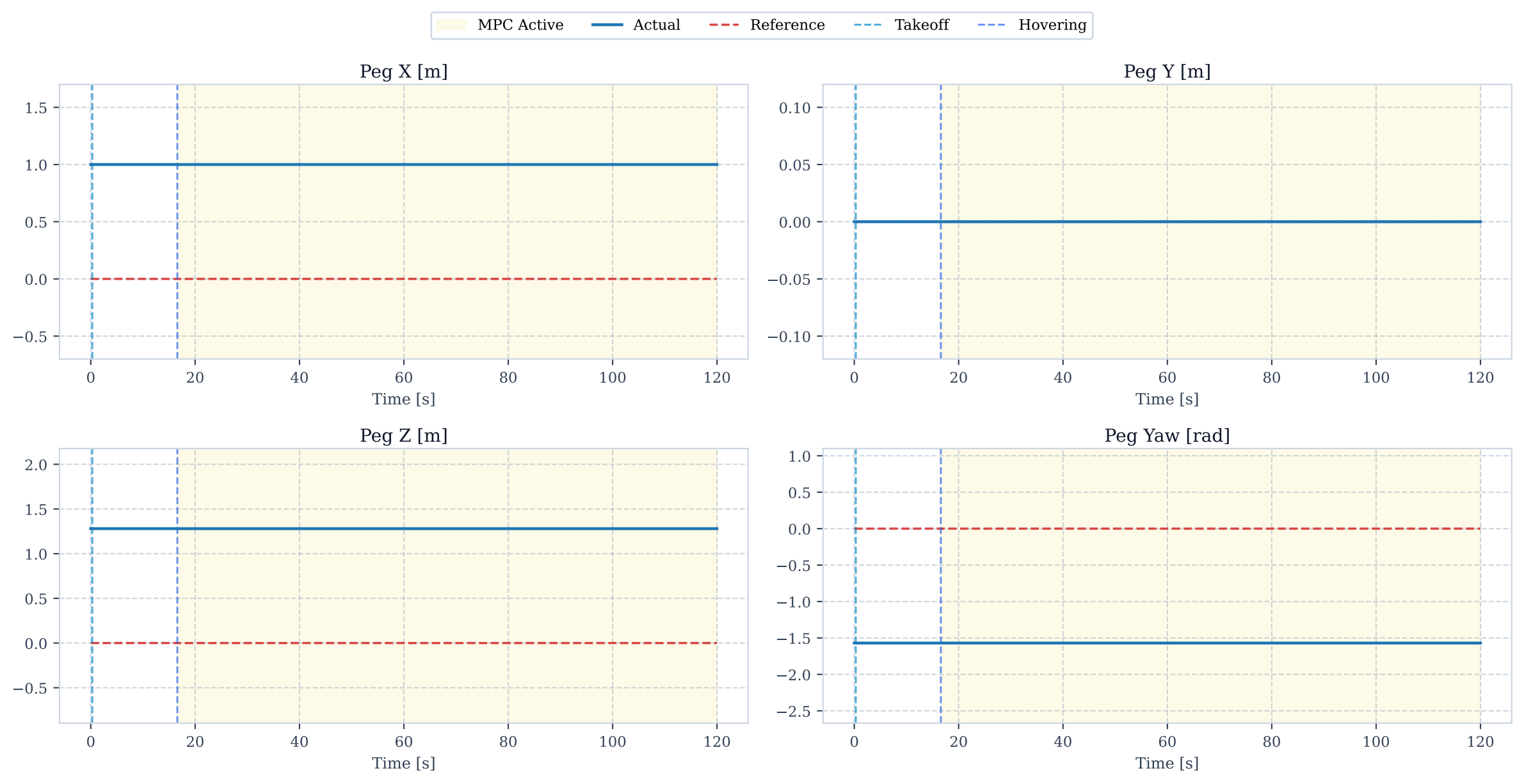

MPC Active
Haptic GuarDrone
Actual
Ref
Takeoff
Hovering
Linear Acc X [m/s^2]
Linear Acc Y [m/s^2]
Linear Acc Z [m/s^2]
Angular Acc X [rad/s^2]
Angular Acc Y [rad/s^2]
Angular Acc Z [rad/s^2]
Time [s]

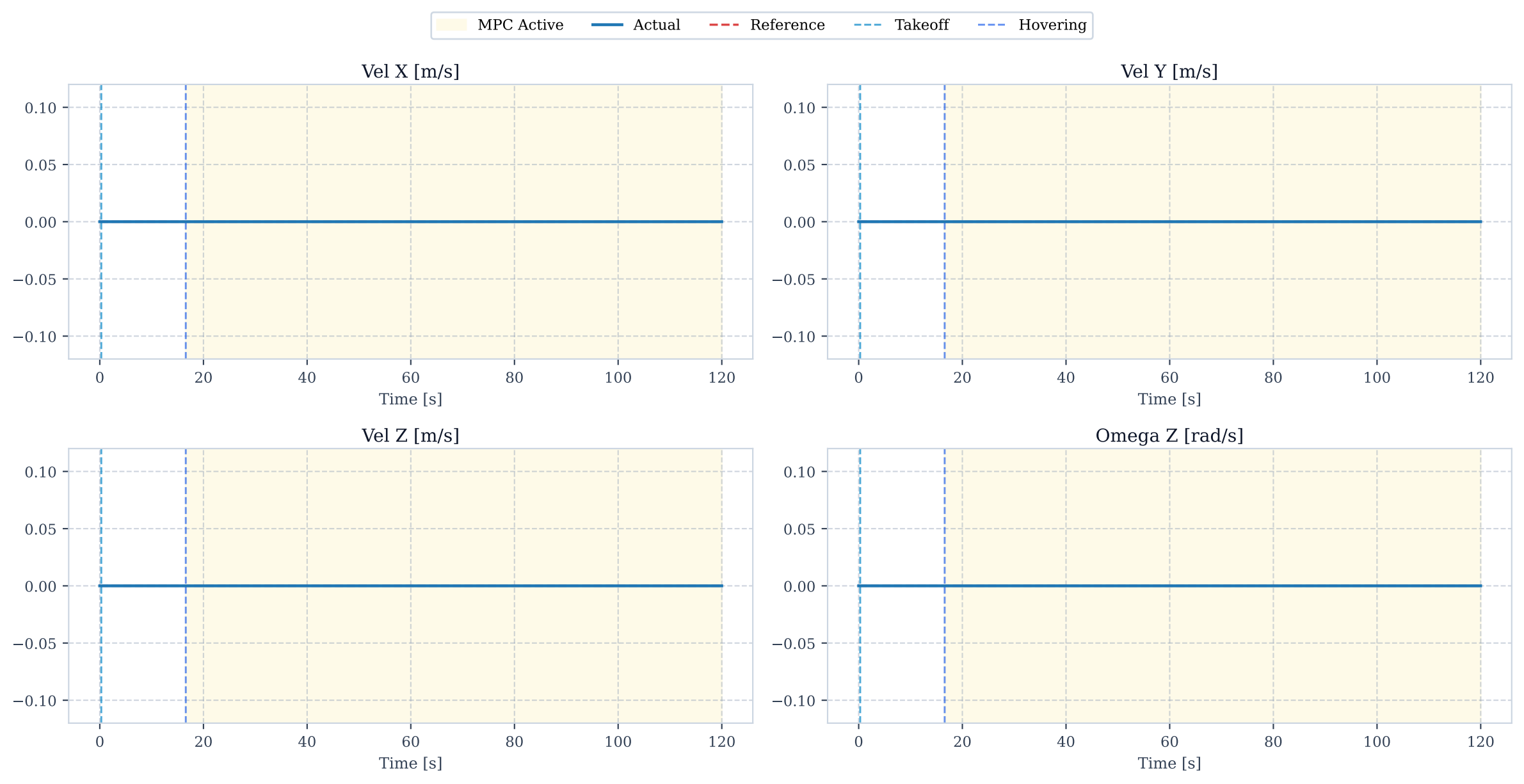

MPC Active
Actual
Ref
Takeoff
Hovering
Force X (Sensor) [N]
Force Y (Sensor) [N]
Force Z (Sensor) [N]
0.10
0.05
0.00
−0.05
−0.10
0
20
40
60
80
100
120
Time [s]

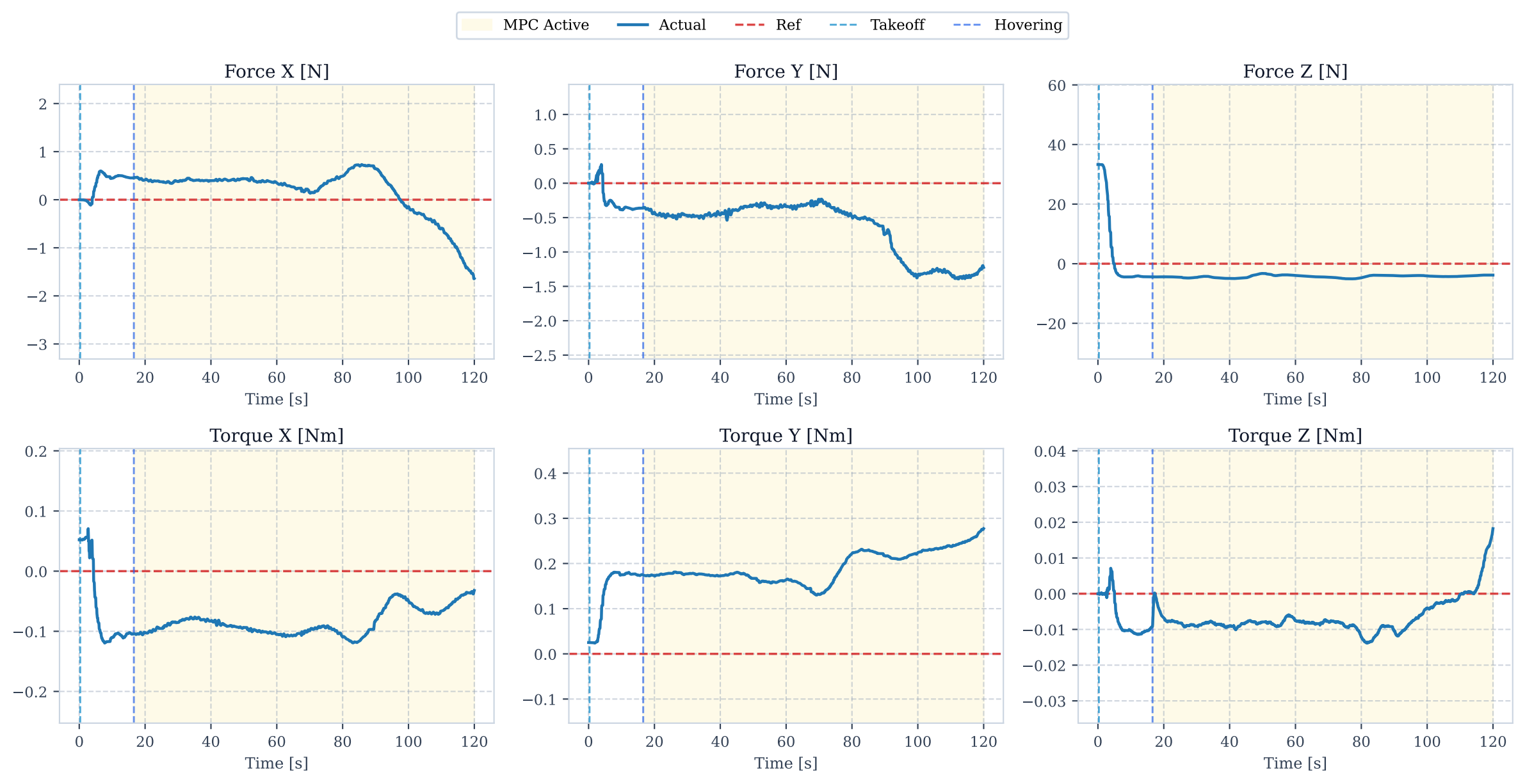

MPC Active
Actual
Ref
Takeoff
Hovering
Δp_x [m] (ENU)
Δp_y [m] (ENU)
Δp_z [m] (ENU)
$\|\Delta p\|$ [m]
0.10
0.05
0.00
−0.05
−0.10
0
20
40
60
80
100
120
Time [s]

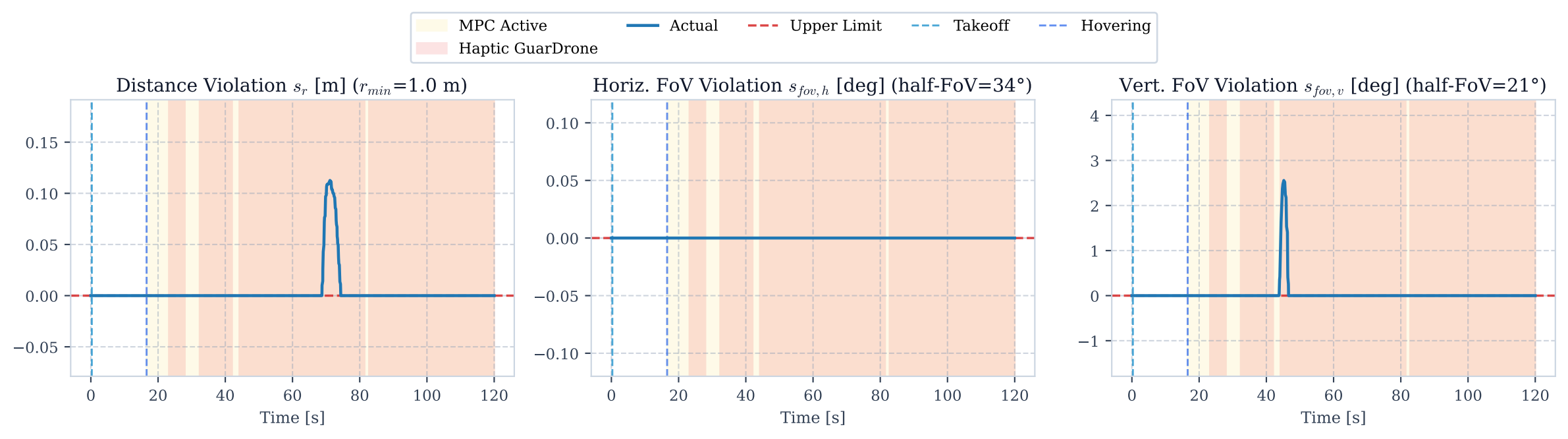

MPC Active
Actual
Ref
Takeoff
Hovering
Δp_{sx} [m] (Sensor X)
Δp_{sy} [m] (Sensor Y)
Δp_{sz} [m] (Sensor Z)
$\|\Delta p_s\|$ [m]
0.10
0.05
0.00
−0.05
−0.10
0
20
40
60
80
100
120
Time [s]

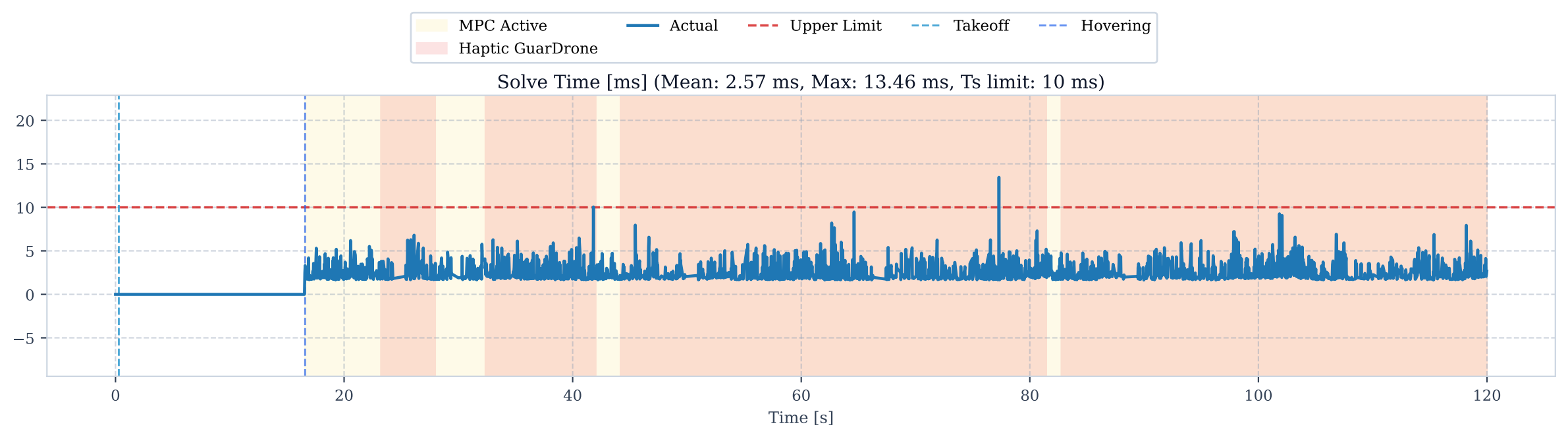

Contact Force Norm
$\|F_{ext}\|$ [N]
$\|F_{ext}\|$ [N]
0
1
2
3
4
5
6
Admittance Displacement Norm
$\|\Delta p\|$ [m]
$\|\Delta p\|$ [m]
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0
20
40
60
80
100
120
Time [s]

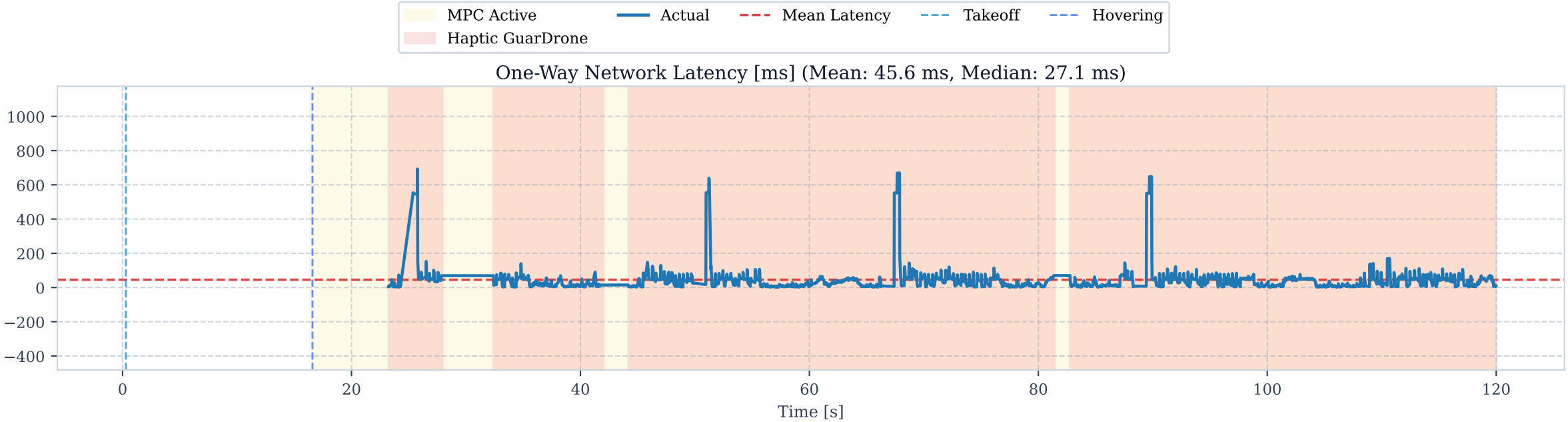

MPC Active
Actual
Reference
Takeoff
Hovering
Peg X [m]
Peg Y [m]
Peg Z [m]
Peg Yaw [rad]
Time [s]
Time [s]
Time [s]
Time [s]


MPC Active
Actual
Reference
Takeoff
Hovering
Vel X [m/s]
Vel Y [m/s]
Vel Z [m/s]
Omega Z [rad/s]
0.10
0.05
0.00
−0.05
−0.10
0
20
40
60
80
100
120
Time [s]


MPC Active
Actual
Ref
Takeoff
Hovering
Force X [N]
Force Y [N]
Force Z [N]
Torque X [Nm]
Torque Y [Nm]
Torque Z [Nm]
Time [s]


MPC Active
Haptic GuarDrone
Actual
Upper Limit
Takeoff
Hovering
Distance Violation s_r [m] (r_{min}=1.0 m)
Horiz. FoV Violation $s_{fov,h}$ [deg] (half-FoV=34°)
Vert. FoV Violation $s_{fov,v}$ [deg] (half-FoV=21°)
Time [s]
Time [s]
Time [s]


MPC Active
Haptic GuarDrone
Actual
Upper Limit
Takeoff
Hovering
Solve Time [ms] (Mean: 2.57 ms, Max: 13.46 ms, Ts limit: 10 ms)
20
15
10
5
0
−5
0
20
40
60
80
100
120
Time [s]


MPC Active
Haptic GuarDrone
Actual
Mean Latency
Takeoff
Hovering
One-Way Network Latency [ms] (Mean: 45.6 ms, Median: 27.1 ms)
1000
800
600
400
200
0
−200
−400
0
20
40
60
80
100
120
Time [s]